

UPENDRA SINGH TOMAR

Birla Nagar, Gwalior, Madhya Pradesh, 474003

+91 887-123-2117 | upendratomar1100@gmail.com | linkedin.com/in/upendra-singh-tomar | github.com/upendra-sin1100
| upendra-singh-tomar.vercel.app

SUMMARY

- Full-stack Data Science student building end-to-end ML applications with Python, TensorFlow, Scikit-Learn, and React.js — spanning healthcare analytics, NLP, and generative AI, with published research in AI-driven sustainability. Focused on machine learning engineering and intelligent data systems.

EDUCATION

Institute of Technology and Management

B.Tech in Data Science (3rd Year, 6th Semester)

Sep. 2023 – May 2027

Gwalior, Madhya Pradesh

RELEVANT COURSEWORK

- Programming in Python, Data Structures & Algorithms, Machine Learning, Deep Learning, Data Science, Artificial Intelligence

PROJECTS

Disease X Command Center | Streamlit, PostgreSQL, Supabase, Machine Learning | [Live Demo]

August 2025

- Built a real-time full-stack hospital management dashboard using Streamlit and Supabase for live data ingestion and patient triage, flagging critical cases and delivering actionable insights to reduce manual triage effort.
- Integrated PostgreSQL and Supabase to manage structured patient data and enable real-time dashboard updates.

Resume Analyzer | Python, Streamlit, Scikit-Learn, K-Means, TF-IDF | [Live Demo]

December 2025

- Built an unsupervised ML pipeline using TF-IDF vectorization and MiniBatch K-Means – chosen for its lower memory footprint and faster convergence over standard K-Means – to cluster unlabeled resume entries into 7 job categories.
- Designed a confidence-based fallback: resumes where cluster-assignment confidence falls below a set threshold are routed to a rule-based keyword classifier instead, improving robustness on ambiguous or edge-case resumes; deployed with real-time PDF parsing on Streamlit.

UpFeed — News Aggregator | React.js, GNews API, NewsAPI, Vercel | [Live Demo]

May 2026

- Developed a responsive news aggregator integrating GNews API and NewsAPI to fetch live articles across multiple categories and sources in real time.
- Built a React.js frontend with reusable components and category-based filtering for efficient news discovery.

ItihaasAI – Indian History Chatbot | Python, RAG, Semantic Search, LLM | [GitHub]

June 2026

- Fine-tuned Llama 3.2 3B Instruct using Unsloth (QLoRA, 4-bit quantization) on Indian history datasets, publishing the adapter to Hugging Face Hub.
- Built an AI-powered chatbot with semantic search and a streaming chat interface, using prompt engineering and a Retrieval-Augmented Generation (RAG) architecture to ground responses in historical context.

AI IDE – Browser-Based Code Editor | React.js, FastAPI, Monaco, Groq, Gemini | [GitHub]

July 2026

- Developed a web-based coding environment inspired by modern IDEs, offering AI-assisted debugging, intelligent code generation, and real-time terminal interaction.
- Built with a React + Vite frontend and a FastAPI backend exposing RESTful APIs, using Monaco Editor with Copilot-style inline suggestions and multi-model AI support via Groq and Gemini.

TECHNICAL SKILLS

Languages: Python, JavaScript

Libraries / Frameworks: React.js, FastAPI, TensorFlow, Streamlit

ML / DL Tools: Scikit-Learn, Keras, Hugging Face Transformers, Pandas, NumPy, Matplotlib, Seaborn

Databases & Cloud: PostgreSQL, Supabase, AWS, Vercel, GitHub

Concepts: REST APIs, Exploratory Data Analysis (EDA), Unsupervised Learning, Prompt Engineering, RAG

PUBLICATIONS

Data-Driven Sustainability in Pharmaceutical Logistics | IIP Series Vol. 6

- Explored cognitive decision intelligence frameworks to reduce medical waste in pharmaceutical supply chains. [Link]

Inverse Design: Generative AI for Carbon Capture Materials (MOFs) | IJAMRED

- Investigated generative AI approaches for designing Metal-Organic Frameworks optimized for carbon capture applications. [Link]

EXTRACURRICULAR

Theater | RGPV University, Central Zone AIU

2023 – 2025

- Performed in 4 major productions for live audiences of 1,000+ and represented RGPV University in Central Zone AIU inter-university theater competitions.